AWS Certified AI Practitioner

AIF-C01 — Quick Revision Cheat Sheet

EXAM AT A GLANCE

Cost	$100 USD
Format	65 Qs (50 scored + 15 unscored), 90 min
Pass score	700/1000 (compensatory)
Validity	3 years
Domain weights	D1: 20% | D2: 24% | D3: 28% | D4: 14% | D5: 14%

DOMAIN 1: Fundamentals of AI & ML (20%)

AI Hierarchy

AI > ML > Deep Learning > Generative AI. Agentic AI = AI that plans, reasons, takes actions.

Types of ML

- **Supervised** (labeled): classification, regression — Comprehend, Rekognition, Fraud Detector
- **Unsupervised** (no labels): clustering, anomaly detection — K-Means, Random Cut Forest
- **Reinforcement** (rewards): DeepRacer, SageMaker RL
- **Self-supervised**: how foundation models are pre-trained

Inference Types (HIGH YIELD)

Type	When to use
Real-time	Low latency, persistent endpoint, always-on traffic
Serverless	Intermittent/unpredictable traffic, scales to zero
Asynchronous	Large payloads (up to 1 GB), long processing (up to 1 hr)
Batch Transform	Bulk offline jobs, no endpoint, scheduled

Key Terms

Overfitting: high train, low test. **Underfitting**: low both. **Bias**: systematic error. **Hyperparameters**: set before training. **Parameters**: learned (weights).

DOMAIN 2: Fundamentals of Generative AI (24%)

Foundation Models on Bedrock

Anthropic Claude, Amazon Titan/Nova, Meta Llama, Mistral, Cohere, AI21, Stability AI

Bedrock vs SageMaker JumpStart (CLASSIC TRAP)

Bedrock	SageMaker JumpStart
Fully managed, serverless API	You deploy on YOUR SageMaker infra
No infrastructure to manage	Full control over instances, hyperparams
Use when: fast, hands-off	Use when: deep customization needed

Amazon Q Variants

- **Q Business**: enterprise data assistant

- **Q Developer**: code assistant in IDEs (was CodeWhisperer)
- **Q in QuickSight**: BI / analytics natural language
- **Q in Connect**: contact center agent assist

Vector Databases on AWS

OpenSearch Serverless (default for Bedrock KB), Aurora PostgreSQL (pgvector), Neptune Analytics, MemoryDB, DocumentDB. **NOT DynamoDB.**

Bedrock Pricing

- **On-demand**: pay per token, variable workloads
- **Provisioned throughput**: high & predictable volume
- **Batch inference**: 50% cheaper for non-urgent jobs

DOMAIN 3: Applications of Foundation Models (28%)

Customization Spectrum (cheapest → expensive)

Method	Use when
Prompt engineering	Simple tasks, no special data
RAG	Private/recent factual data, reduce hallucinations
Fine-tuning (instruction)	Specific style/format/tone, labeled examples
Domain adaptation FT	Limited domain-specific labeled data
Continued pre-training	Lots of UNLABELED domain data

Memory aid: Style/tone → Fine-tuning. Knowledge/facts → RAG. Domain expertise + unlabeled corpus → Continued pre-training.

Prompt Engineering Techniques

- **Zero-shot**: no examples
- **Few-shot / single-shot**: provide examples
- **Chain-of-thought (CoT)**: "think step by step" — boosts reasoning

Prompt Risks

Prompt injection, prompt leaking, jailbreaking, poisoning, hijacking. Mitigate with **Bedrock Guardrails**.

Evaluation Metrics

- **BLEU** → Translation
- **ROUGE** → Summarization
- **BERTScore** → Semantic similarity
- **Perplexity** → How well LM predicts (lower=better)
- **Accuracy / Precision / Recall / F1** → Classification

Inference Parameters

Temperature: 0=deterministic, 1=creative. **Top-p / Top-k**: sampling controls. **Max tokens**: caps length. **Stop sequences**: halt generation.

Bedrock Agents

FM + Action Groups (Lambda/APIs) + Knowledge Bases + Memory. Use for multi-step tasks needing actions (book, query, execute).

DOMAIN 4: Guidelines for Responsible AI (14%)

Responsible AI Principles

Fairness, Explainability, Privacy & Security, Safety, Controllability, Veracity & Robustness, Governance, Transparency.

Transparency vs Explainability (TRAP)

- **Transparency**: openness about system — data, design, limits (documentation)
- **Explainability**: WHY did the model make THIS prediction? (feature attribution)

AWS Responsible AI Services

- **SageMaker Clarify**: bias detection + explainability (SHAP)
- **Bedrock Guardrails**: content filters, denied topics, PII redaction, grounding
- **AWS AI Service Cards**: docs for AWS-managed AI services
- **SageMaker Model Cards**: docs for YOUR custom models
- **Amazon A2I**: human-in-the-loop review for low-confidence
- **SageMaker Model Monitor**: data drift & concept drift detection

DOMAIN 5: Security, Compliance & Governance (14%)

Core Security Services

- **IAM**: least privilege, roles, IAM Identity Center for Q Business
- **KMS**: encryption at rest (S3, EBS, SageMaker)
- **VPC Endpoints (PrivateLink)**: keep Bedrock/SageMaker traffic private
- **Macie**: detect PII in S3
- **CloudTrail**: API audit logs
- **Bedrock Model Invocation Logging**: log prompts & responses

Bedrock Data Privacy (KEY FACT)

> Customer prompts and outputs are **NOT** used to train base FMs. Data stays yours. Bedrock is HIPAA-eligible, SOC, GDPR-ready.

GenAI Security Scoping Matrix

Scope 1: Consumer apps | Scope 2: Enterprise SaaS | Scope 3: Pre-trained FMs (Bedrock on-demand) | Scope 4: Fine-tuned | Scope 5: Self-trained. Higher scope = more control + responsibility.

Compliance Tools

AWS Artifact (compliance reports), Audit Manager, AWS Config, SCPs (org-wide), AWS Budgets (cost control).

FINAL EXAM-DAY TIPS

1. Read questions twice — keywords like "private data", "no infrastructure", "unlabeled" decide the answer.
2. **RAG** is usually right for grounding/private data. **Fine-tuning** for style/tone.
3. **Bedrock** = managed. **SageMaker JumpStart** = you manage.
4. **OpenSearch Serverless** for vector storage (NOT DynamoDB).
5. No penalty for guessing — answer every question.
6. Flag & review uncertain questions; trust your first instinct.